

PRODUCT LITERATURE

BUTEXINE SYRUP

Each 5 ml contains:

Bromhexine Hydrochloride	4.0 mg
Salbutamol Sulphate	2.4 mg

Preservatives

Methyl Paraben	0.1%w/v
Propyl Paraben	0.01%w/v

Description

Reddish orange colour clear syrup with minty odour and sweet taste.

Pharmacodynamics

As a beta-adrenergic stimulant for relief of bronchospasm such as occurs with asthma, bronchitis, emphysema. It has a highly selective action on the receptors in bronchial muscle and in therapeutic dosage, little or no action on the cardiac receptors.

Pharmacokinetic

Salbutamol is readily absorbed from the gastro-intestinal tract and is subject to first pass metabolism in the liver. Peak plasma concentrations occur within one to four hours after oral administration. After multiple oral doses of salbutamol 4mg four times a day, steady-state plasma concentrations are obtained after 3 days. About half is excreted in the urine as an inactive sulphate conjugate following oral administration. The bioavailability of orally administered salbutamol is about 50%

Indication

A bronchodilator with mucolytic expectorant for clearance of viscid purulent sputum in acute and chronic bronchitis and complicating conditions, bronchial asthma, bronchiectasis and whooping cough.

Route of administration

Oral.

Recommended Dose

Children below 6 years : 1.25 to 2.5ml

Older Children : 2.5ml to 5.0ml

Adult: : 5ml

To be given 3 to 4 times a day or as directed by the Physician.

Contraindication

Although intravenous salbutamol and occasionally salbutamol syrup are used in the management of uncomplicated premature labour, salbutamol presentations should not be used for threatened abortion during the first or second trimester of pregnancy. Should not be used in patients hypersensitive to any of the product ingredients.

Warning and Precautions

Butexine Syrup may cause severe allergy and serious skin reactions. Stop using Butexine Syrup and seek medical assistance immediately if you experience any of the following symptoms:

- 1) severe allergy: breathing difficulties, light headedness, skin swellings or rash
- 2) severe skin reaction: skin reddening, blisters, rash, fever, sore throat or eye irritation

Very rare cases of chronically associated severe skin impairments such as Stevens Johnson Syndrome, Toxic Epidermal Necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP) have been reported. In most cases, these could be explained by the severity of the underlying disease or concomitant administration of another drug. In the early stages of such severe skin reactions, initially only nonspecific flu-like symptoms appear, e.g. fever, arthralgia, runny nose, cough, and sore throat. If skin or mucous membrane damage occurs, seek medical advice immediately and discontinue treatment as a precaution.

Bronchodilators should not be the only or main treatment in patients with severe or unstable asthma. Severe asthma requires regular medical assessment including lung function testing as patients are at risk of severe attacks and even death. Physicians should consider using oral corticosteroid therapy and/or the maximum recommended dose of inhaled corticosteroid in those patients. Patients should seek medical advice if treatment with Salbutamol Syrup 2mg/5ml becomes less effective. The dosage or frequency of administration should only be increased on medical advice. Patients taking Salbutamol Syrup 2mg/5ml may also be receiving short-acting inhaled bronchodilators to relieve symptoms. Increasing use of bronchodilators in particular short-acting inhaled beta₂- agonists to relieve symptoms indicates deterioration of asthma control. The patient should be instructed to seek medical advice if short-acting relief bronchodilator treatment becomes less effective or they need more inhalations than usual. In this situation patients should be reassessed and consideration given to the need for increased anti-inflammatory therapy (eg. Higher doses of inhaled corticosteroids or a course of oral corticosteroid). Severe exacerbations of asthma must be treated in the normal way.

Patients should be warned that if either the usual relief is diminished or the usual duration of action is reduced, they should not increase the dose or its frequency of administration, but should seek medical advice. Salbutamol causes peripheral vasodilation which may result in reflex tachycardia and increased cardiac output. Caution should be used in patients suffering from angina, severe tachycardia or thyrotoxicosis. Cardiovascular effects may be seen with sympathomimetic drugs, including salbutamol. There is some evidence from post-marketing data and published literature of rare occurrences of myocardial ischaemia associated with salbutamol. Patients with underlying severe heart disease (e.g. ischaemic heart disease, arrhythmia or severe heart failure) who are receiving salbutamol should be warned to seek medical advice if they experience chest pain or other symptoms of worsening heart disease. Attention should be paid to assessment of symptoms such as dyspnoea and chest pain, as they may be of either respiratory or cardiac origin. Severe exacerbations of asthma must be treated in the usual manner. Caution should be exercised in its use with anaesthetic agents such as chloroform, cyclopropane, halothane and other halogenated agents. Salbutamol should not cause difficulty in micturition (urination) because unlike sympathomimetic drugs such as ephedrine, it does not stimulate α -adrenoceptors. However, there have been reports of difficulty in micturition in patients with prostatic enlargement. Salbutamol should only be used during pregnancy if considered essential by the physician. Salbutamol does not contain sugars. This product should not be diluted. Potentially serious hypokalaemia may result from beta-2 agonist therapy mainly from parenteral and nebulised administration. Particular caution is advised in acute severe

asthma as this effect may be potentiated by hypoxia and by concomitant treatment with xanthine derivatives, steroids. It is recommended that serum potassium levels are monitored in such situations. In common with other β -adrenoceptor agonists, salbutamol can induce reversible metabolic changes such as increased blood glucose levels. Diabetic patients may be unable to compensate for the increase in blood glucose and the development of ketoacidosis has been reported. Concurrent administration of corticosteroids can exaggerate this effect. This medicinal product contains small amounts of ethanol (alcohol), less than 100mg per 5ml dose.

Tocolysis: Serious adverse reactions including death have been reported after administration of terbutaline/ salbutamol to women in labor. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

Interactions with Other Medicaments

Caution should be exercised during use with anaesthetic agents such as chloroform, cyclopropane, halothane and other halogenated agents. The effects of this product may be altered by guanethidine, reserpine, methyl dopa, tricyclic antidepressants. Salbutamol oral preparations and non-selective beta-blocking drugs, such as propranolol should not usually be prescribed together. Salbutamol is not contraindicated in patients under treatment with monoamine oxidase inhibitors (MAOIs).

Pregnancy and Lactation

Pregnancy

Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus. As with the majority of drugs, there is little published evidence of its safety in the early stages of human pregnancy, but in animal studies there was evidence of some harmful effects on the foetus at very high dose levels.

Breastfeeding

As salbutamol is probably secreted in breast milk its use in nursing mothers requires careful consideration. It is not known whether salbutamol has a harmful effect on the neonate, and so its use should be restricted to situations where it is felt that the expected benefit to the mother is likely to outweigh any potential risk to the neonate.

Fertility

There is no information on the effects of salbutamol on human fertility. There were no adverse effects on fertility in animals.

Side Effects

Immune System Disorders

Frequency not known: Anaphylactic reactions including anaphylactic shock.

Skin and Subcutaneous Skin Disorders

Frequency not known: Severe skin reactions (including Stevens Johnson syndrome, Toxic epidermal necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP).

Summary of the safety profile.

The most common side effect of Salbutamol Syrup 2mg/5ml is fine tremor of the hands, which may interfere with precise manual work. Tension, restlessness and a rapid heartbeat may also occur. There have been very rare reports of muscle cramps. Hypersensitivity reactions such as angioedema, urticaria, bronchospasm, hypotension and collapse have rarely been reported. Potentially serious hypokalaemia may result from β_2 -agonist therapy. Occasional headaches have also been reported. As with other drugs in this class rare reports of hyperactivity in children have been reported.

Symptoms and Treatment of Overdose

Symptoms.

The most common signs and symptoms of overdose with salbutamol are transient beta agonist pharmacologically mediated events, including tachycardia, tremor, hyperactivity and metabolic effects including hypokalaemia. Salbutamol overdose may lead to Hypokalaemia (abnormally low potassium concentration in the blood). Serum potassium levels should therefore be monitored.

Lactic acidosis has been reported in association with high therapeutic doses as well as overdoses of short-acting beta-agonist therapy, therefore monitoring for elevated serum lactate and consequent metabolic acidosis (particularly if there is persistence or worsening of tachypnea despite resolution of other signs of bronchospasm such as wheezing) may be indicated in the setting of overdose. Nausea, vomiting and hyperglycaemia have been reported, predominantly in children and when salbutamol overdose has been taken via the oral route.

Treatment

The preferred antidote for overdose with salbutamol sulphate is a cardio selective beta-blocking agent, which should be used with caution in patients with a history of bronchospasm. Further management should be as clinically indicated or as recommended by the national poisons center, where available.

Packing

Packed in plastic container of 60 ml, 100 ml and 120 ml.

Storage Conditions

Store below 30°C in a dry place, protected from direct light. Keep away from reach of children.

SHAKE WELL BEFORE USE

Expiry Date

2 years from date of manufacture.

Product Registration Holder / Distributor:

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Lot 10 & 11, PERDA Industrial Park,
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Malaysian Drug Registration No. :

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